

JC2002 Java Programming

Introduction

September 2026

JC2002 Java Programming

Day 1, Session 1: Course introduction and practicalities

Icebreaker activities

- Who am I?
- How familiar are you with Java and programming in general?
- What do you expect from this course?

Course arrangements

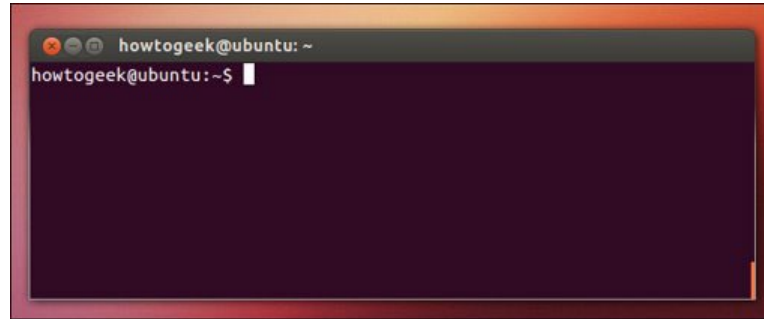
- **Lectures** (theory sessions)
- **Practical sessions**
- **Practical assignment**, 30% of the assessment
 - Assignment will be released in the 10th week
 - Submission in MyAberdeen (deadline: December 8)
- **Exam**, 70% of the assessment
 - Paper-based closed-book exam

Course topics

- **Week 1-3:** Java fundamentals
 - Using the basic Java tools, version control, testing
 - Java syntax, conditional structures, loops, classes, objects, methods
- **Week 4-6:** Object oriented programming, graphical user interfaces
 - OOP principles: class inheritance, polymorphism, abstract classes
 - Using Swing for implementing graphical user interfaces in Java
- **Week 7-9:** Concurrency and data structures
 - Multithreading and memory concepts
 - String operations, collections
- **Week 10-12:** Advanced topics in Java
 - File handling, database programming, etc.

Course topics – not only Java

- Professional software development practices
 - Version source control to keep track of development
 - Use of testing, and automated testing to ease development
 - Linux command line, read documentation, etc.



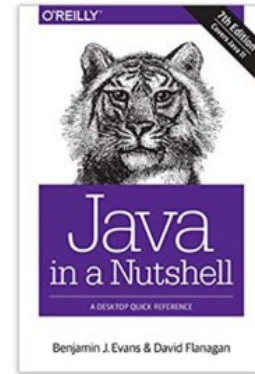
Read the course textbooks

Lectures are only part of the process. You also need to read about the language!

Essential (2)▼

 **E-BOOK Java: how to program: early objects** ✓
Deitel, Harvey, 11th, Pearson Education, 2018
Essential Key resource - please purchase e-copy 
Complete [Check availability >](#)

 **E-BOOK Head first Java: a brain friendly guide** ✓
Sierra, Kathy, 2nd, O'Reilly Media, 2005
Note: There are also a few paper copies of this book in the library.
Essential Key resource - please purchase e-copy 
Complete Available at Sir Duncan Rice Library Floor 7 : 005.133 J SIE



- Course materials in library and online (We'll be adding more)
- Lots of materials in the library on software development
-> both as books and online books

Assignment

- Programming assignment – 30% of the mark
- Exam – 70% of the mark
- Late submission policy
 - up to 24 hours late will incur a 10% penalty to your mark
 - up to 1 week late will incur a 25% penalty
 - Any hand-ins after that will be classified as no submission.
- Feedback
 - Feedback will be given no later than three weeks after you submit your work

Academic integrity

- **Plagiarism** is the unattributed use of other people's work or ideas, in work submitted for assessment.
- **Collusion** occurs where there is unauthorised collaboration between students in the writing of an individual assignment.
- **"Contract cheating"** is a form of academic misconduct where a student submits work that has been produced by someone other than the student, whether this has been paid for or not.

See: <https://www.abdn.ac.uk/students/academic-life/academic-integrity.php>

Academic integrity

- Plagiarism

Reference and cite appropriately sources used in assessments and other learning activities.

Include direct quotes, if they are deemed appropriate by your subject area, acknowledged with quotation marks or indented, and remember to reference or cite your sources.

Paraphrase ideas from your sources using your own words, but always reference or cite the original author(s).

Resist making only minor changes to the original text and using them in assignments.

Do not copy and paste into your work sections of material from the Internet, books, papers, reports, or other published work.

Do not self-plagiarise by re-using previous marked assessments and re-submitting them.

Reference or cite the sources of tables, figures, diagrams, and images used in assignments.

- Collusion

Do not collude with others when writing your own individual assignment.

Create a study plan and timetable for assessments, to avoid the pressure to collude.

Do not allow your peers to access, copy, or submit your assignment as their own work.

- “Contract cheating”

Never commission assignment outlines or plans from online sources.

Do not request anyone, including friends, family, or tutors to prepare an assignment for you, whether paid or unpaid.

Never engage with ‘essay mills’ or custom sites to prepare assessments of any kind.

Plagiarism in reports

- If taking text verbatim, put it in quotes and *cite the source*
- Alternatively, paraphrase or summarize and *cite the source*
- As a rule of thumb, don't do this for more than 3-4 sentences
- See <https://integrity.mit.edu/handbook/writing-original-work> for an excellent guide

Plagiarism in source code

- An excellent guide: <https://integrity.mit.edu/handbook/writing-code>
 - Never copy another student's code
 - Don't simply reuse code
- Cite the sources of code that you use within your code
- Ensure that any licensing conditions are adhered to

Automatic plagiarism detection

- Similarity checking tools spot modifications, such as:
 - Change variable, constant or function/procedure names
 - Simply replace language constructs
 - Change indentation or separation within source code
 - See http://www.upenn.edu/academicintegrity/ai_computercode.html for further examples



In this course, plagiarism detection tool
will be used to check the coursework assignments!

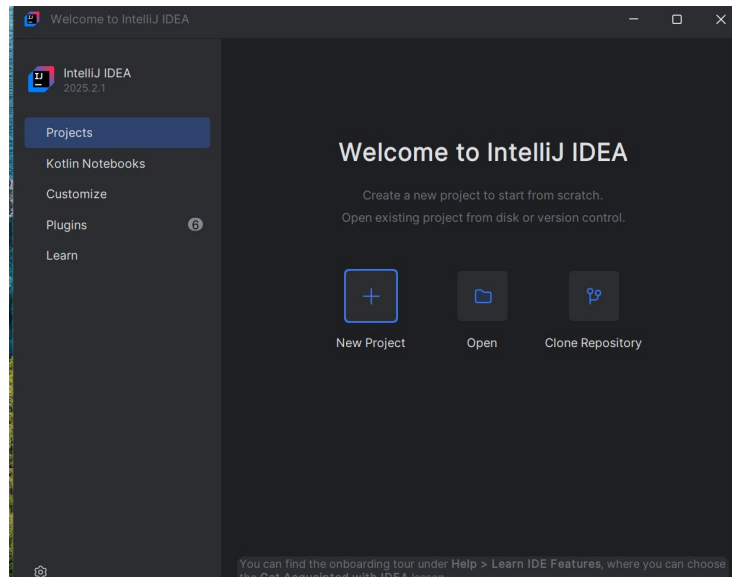
Ask for help if you are stuck

- If something is unclear, then ask someone or look it up
- If you still find it hard, get in touch with teacher
- Copying something from someone else, or via the internet, will not help you learn how to do the work

It is ALWAYS better to submit a poor assignment of your work with comments about what you tried, and struggle with, than to submit someone else's work.

Suggested Environment: IntelliJ IDEA Community

- Please download the Community Edition to your own machine
- <https://www.jetbrains.com.cn/idea/>



Version control

- Keeping code on local machines only is unsafe (disk failure)
- It is hard to work as a team if the members are working on their own versions of the code
- The solution: ***version control***
 - Version control allows to make remote copies of your code for sharing the code and rolling back to earlier versions easily
 - In this course, we will use **Git** for version control and as way to start working with remote repositories, such as **GitHub**

Questions, comments?